

Bachelor of Science (Honours) in Data Science and Artificial Intelligence

**DA 102 – Basics of Spread Sheet Modeling**

# Learning Objectives



- 01** Examples on Descriptive Analytics
- 02** Absolute addressing method of cells
- 03** Elementary predictive analytics example
- 04** Elementary analysis of demand curve estimation
- 05** Range names





# **Descriptive Analysis - Employees Weekly Wages**



# Employees Weekly Wages



- **The file contains four columns**
  - Employee Name
  - Number of hours worked
  - Wage per hour to be paid
- **The task is to compute**
  - Weekly salary to be paid to each employee
  - Total weekly salary paid
  - Average weekly salary paid
  - This is an example of descriptive analytics
- **This problems involves writing simple formula, copy the formula in several cells. Use mathematical functions SUM, AVERAGE**

# Employees Weekly Wages



- **Formula 1: For every employee use the following formula to compute Weekly salary:**
  - Wage Per Hour \* Number of hours worked
  - Write this formula against one employee
  - Copy the formula all the other employees
- **Formula 2: To compute total salary paid, use SUM function**
- **Formula 3: To compute average salary, use AVERAGE function**
- **Inputs: Columns C and D**
- **Outputs: Column E**





# **Descriptive Analysis – Bakery Owner Dues to Suppliers**



# Bakery Owner Dues to Suppliers – Data Model 1



- The file contain inputs
- **Prices**
  - Prices of sugar from supplier 1, supplier 2, ..., supplier 6
  - Prices of butter from supplier 1, supplier 2, ..., supplier 6
  - Prices of flour from supplier 1, supplier 2, ..., supplier 6
- **Quantity Supplied**
  - Quantity of sugar supplied by supplier 1, supplier 2, ..., supplier 6
  - Quantity of butter supplied by supplier 1, supplier 2, ..., supplier 6
  - Quantity of flour supplied by supplier 1, supplier 2, ..., supplier 6

# Bakery Owner Dues to Suppliers



- **Problem:**

- Find the amount to be given to supplier 1 for Sugar
- Find the amount to be given to supplier 1 for Butter
- Find the amount to be given to supplier 1 for Flour
- Find total the amount to be given to supplier 1
  
- Find the amount to be given to supplier 2 for Sugar
- Find the amount to be given to supplier 2 for Butter
- Find the amount to be given to supplier 2 for Flour
- Find total the amount to be given to supplier 2
- ...
- Find the amount to be given to supplier 6 for Sugar
- Find the amount to be given to supplier 6 for Butter
- Find the amount to be given to supplier 6 for Flour
- Find total the amount to be given to supplier 6

# Bakery Owner Dues to Suppliers – Data Model 2



- The file contains inputs
- For each supplier
  - Sugar column contains three sub-columns
    - Price
    - Quantity
    - Total
  - Butter column contains three sub-columns
    - Price
    - Quantity
    - Total
  - Flour column contains three sub-columns
    - Price
    - Quantity
    - Total



# Bakery Owner Dues to Suppliers – Data Model 1

| Prices     |         |          |         |          |
|------------|---------|----------|---------|----------|
|            | Sugar   | Butter   | Flour   |          |
| Supplier 1 | \$ 0.32 | \$ 1.57  | \$ 0.11 |          |
| Supplier 2 | \$ 0.35 | \$ 1.54  | \$ 0.10 |          |
| Supplier 3 | \$ 0.25 | \$ 1.54  | \$ 0.21 |          |
| Supplier 4 | \$ 0.29 | \$ 1.24  | \$ 0.10 |          |
| Supplier 5 | \$ 0.35 | \$ 1.30  | \$ 0.18 |          |
| Supplier 6 | \$ 0.27 | \$ 1.42  | \$ 0.15 |          |
| Quantity   |         |          |         |          |
|            | Sugar   | Butter   | Flour   |          |
| Supplier 1 | 364     | 391      | 220     |          |
| Supplier 2 | 387     | 245      | 314     |          |
| Supplier 3 | 290     | 211      | 200     |          |
| Supplier 4 | 340     | 265      | 330     |          |
| Supplier 5 | 261     | 345      | 246     |          |
| Supplier 6 | 365     | 232      | 390     |          |
| Cost       |         |          |         |          |
|            | Sugar   | Butter   | Flour   | Total    |
| Supplier 1 | 116.48  | 613.87   | 24.20   | 754.55   |
| Supplier 2 | 135.45  | 377.30   | 31.40   | 544.15   |
| Supplier 3 | 72.50   | 324.94   | 42.00   | 439.44   |
| Supplier 4 | 98.60   | 328.60   | 33.00   | 460.20   |
| Supplier 5 | 91.35   | 448.50   | 44.28   | 584.13   |
| Supplier 6 | 98.55   | 329.44   | 58.50   | 486.49   |
| Total      | 612.93  | 2,422.65 | 233.38  | 3,268.96 |

# Bakery Owner Dues to Suppliers – Data Model 2



|            | Sugar   |          |        | Butter  |          |         | Flour   |          |        |             |
|------------|---------|----------|--------|---------|----------|---------|---------|----------|--------|-------------|
|            | Price   | Quantity | Total  | Price   | Quantity | Total   | Price   | Quantity | Total  |             |
| Supplier 1 | \$ 0.32 | 364      | 116.48 | \$ 1.57 | 391      | 613.87  | \$ 0.11 | 220      | 24.2   |             |
| Supplier 2 | \$ 0.35 | 387      | 135.45 | \$ 1.54 | 245      | 377.3   | \$ 0.10 | 314      | 31.4   |             |
| Supplier 3 | \$ 0.25 | 290      | 72.5   | \$ 1.54 | 211      | 324.94  | \$ 0.21 | 200      | 42     |             |
| Supplier 4 | \$ 0.29 | 340      | 98.6   | \$ 1.24 | 265      | 328.6   | \$ 0.10 | 330      | 33     |             |
| Supplier 5 | \$ 0.35 | 261      | 91.35  | \$ 1.30 | 345      | 448.5   | \$ 0.18 | 246      | 44.28  |             |
| Supplier 6 | \$ 0.27 | 365      | 98.55  | \$ 1.42 | 232      | 329.44  | \$ 0.15 | 390      | 58.5   |             |
|            |         |          | 612.93 |         |          | 2422.65 |         |          | 233.38 | \$ 3,268.96 |





# Absolute Addressing



# Bakery Owner Dues to Suppliers



| Prices     |         |         |         |  |
|------------|---------|---------|---------|--|
|            | Sugar   | Butter  | Flour   |  |
| Supplier 1 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 2 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 3 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 4 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 5 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 6 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
|            |         |         |         |  |
| Quantity   |         |         |         |  |
|            | Sugar   | Butter  | Flour   |  |
| Supplier 1 | 364     | 391     | 220     |  |
| Supplier 2 | 387     | 245     | 314     |  |
| Supplier 3 | 290     | 211     | 200     |  |
| Supplier 4 | 340     | 265     | 330     |  |
| Supplier 5 | 261     | 345     | 246     |  |
| Supplier 6 | 365     | 232     | 390     |  |
|            |         |         |         |  |

# Bakery Owner Dues to Suppliers



- The file contains inputs
- Prices
  - Each supplier supplies sugar for same price
  - Each supplier supplies butter for same price
  - Each supplier supplies flour for same price
- Quantity Supplied
  - Quantity of sugar supplied by supplier 1, supplier 2, ..., supplier 6
  - Quantity of butter supplied by supplier 1, supplier 2, ..., supplier 6
  - Quantity of flour supplied by supplier 1, supplier 2, ..., supplier 6

# Bakery Owner Dues to Suppliers



|            | Price  |        |        |
|------------|--------|--------|--------|
|            | \$0.40 | \$1.20 | \$0.12 |
| Quantity   | Sugar  | Butter | Flour  |
| Supplier 1 | 364    | 391    | 220    |
| Supplier 2 | 387    | 245    | 314    |
| Supplier 3 | 290    | 211    | 200    |
| Supplier 4 | 340    | 265    | 330    |
| Supplier 5 | 261    | 345    | 246    |
| Supplier 6 | 365    | 232    | 390    |



# Formula Copying Associated Issue

|     |         |            |            |             |         |           |       |             |   |   |
|-----|---------|------------|------------|-------------|---------|-----------|-------|-------------|---|---|
| D11 |         |            |            |             |         |           |       |             |   |   |
|     | B       | C          | D          | E           | F       | G         | H     | I           | J | K |
| 11  |         |            |            | Price       |         |           |       |             |   |   |
| 12  |         |            |            | \$0.40      | \$1.20  | \$0.12    |       |             |   |   |
| 13  |         |            | Quantity   | Sugar       | Butter  | Flour     |       |             |   |   |
| 14  |         |            | Supplier 1 | 364         | 391     | 220       |       |             |   |   |
| 15  |         |            | Supplier 2 | 387         | 245     | 314       |       |             |   |   |
| 16  |         |            | Supplier 3 | 290         | 211     | 200       |       |             |   |   |
| 17  |         |            | Supplier 4 | 340         | 265     | 330       |       |             |   |   |
| 18  |         |            | Supplier 5 | 261         | 345     | 246       |       |             |   |   |
| 19  |         |            | Supplier 6 | 365         | 232     | 390       |       |             |   |   |
| 20  |         |            |            |             |         |           |       |             |   |   |
| 21  |         |            | Cost       |             |         |           |       |             |   |   |
| 22  |         |            |            | Sugar       | Butter  | Flour     | Total |             |   |   |
| 23  | E14*E12 | Supplier 1 | \$         | 145.60      | \$      | 469.20    | \$    | 26.40       |   |   |
| 24  | E15*E13 | Supplier 2 |            | #VALUE!     | #VALUE! | #VALUE!   |       |             |   |   |
| 25  | E16*E14 | Supplier 3 | \$         | 1,05,560.00 | \$      | 82,501.00 | \$    | 44,000.00   |   |   |
| 26  | E17*E15 | Supplier 4 | \$         | 1,31,580.00 | \$      | 64,925.00 | \$    | 1,03,620.00 |   |   |
| 27  | E18*E16 | Supplier 5 | \$         | 75,690.00   | \$      | 72,795.00 | \$    | 49,200.00   |   |   |
| 28  | E19*E17 | Supplier 6 | \$         | 1,24,100.00 | \$      | 61,480.00 | \$    | 1,28,700.00 |   |   |

Sheet 4 of 5

PageStyle\_Cost 2

English (India)

CountA: 0; Sum: 0

140%



# Formula To Compute Money For Each Item

- Price for sugar by supplier 1 =  $E14 * E12$
- When formula in cell E23 is copied to E24, E25, E26, E27 and E28, we want the follow and not the one in produced by copying formula show in red color
  - Price for sugar by supplier 2 =  $E15 * E12$  (NOT  $E15 * E13$ )
  - Price for sugar by supplier 3 =  $E16 * E12$  (NOT  $E16 * E14$ )
  - Price for sugar by supplier 4 =  $E17 * E12$  (NOT  $E17 * E15$ )
  - Price for sugar by supplier 5 =  $E18 * E12$  (NOT  $E18 * E16$ )
  - Price for sugar by supplier 6 =  $E19 * E12$  (NOT  $E19 * E17$ )
- The requirement of E12 cell should be referred constantly without changing while performing copying is known as absolute addressing
- This is achieved by writing the formula as:  $E14 * \$E\$12$ .
- Preceding \$ in cell referencing make Excel not to change the value of column name AND row number of E12 while performing copying.



# Absolute Reference & Formula Copying

- Write the following formula for price of sugar supplied by supplier 1 =  $E14 * \$E\$12$
- When formula in cell E23 is copied to E24, E25, E26, E27 and E28, the following content will be copied in all cells
  - Price for sugar by supplier 2 =  $E15 * \$E\$12$
  - Price for sugar by supplier 2 =  $E16 * \$E\$12$
  - Price for sugar by supplier 3 =  $E17 * \$E\$12$
  - Price for sugar by supplier 4 =  $E18 * \$E\$12$
  - Price for sugar by supplier 5 =  $E19 * \$E\$12$
- Cell referencing using notation  $\$E\$12$  is known as absolute referencing.
- Copying will not change this reference to cell E12 when referred as  $\$E\$12$ .



## **Absolute Addressing – Fixed row, varying column**





# Fixed Row, Varying Column and Effect of Copy

- Cell is referred as (Column Name, Row Number).
- In order for copy not to change row number, place DOLLAR sign in front of row.
- Example: Cell B1 contains the formula A\$10
- When formula in cell B1 is copied to cell C1,
  - As there is NO DOLLAR sign in column name, it is varied to B.
  - As there is DOLLAR sign in row 10, the row number 10 remains constant.
  - There the formula produced in cell C1 is: B10



## **Absolute Addressing – Varying row, fixed column**





# Fixed Row, Varying Column and Effect of Copy

- Cell is referred as (Column Name, Row Number).
- In order for copy not to change column name, place DOLLAR sign in front column name.
- Example: Cell B1 contains the formula \$A10
- When formula in cell B1 is copied to cell B2,
  - As there is DOLLAR sign in column name, column name is NOT varied. It is A
  - As there is NO DOLLAR sign in row 10, the row number changes to 11.
  - There the formula produced in cell B2 is: A11



## **Absolute Addressing – fixed row, fixed column**



# Fixed Row, Varying Column and Effect of Copy



- Cell is referred as (Column Name, Row Number).
- In order for copy not to change column name, place DOLLAR sign in front column name.
- In order for copy not to change row number, place DOLLAR sign in front row number
- Example: Cell B1 contains the formula `$A$10`
- When formula in cell B1 is copied to cell D2,
  - As there is DOLLAR sign in column name, column name is NOT varied. It remains A.
  - As there is DOLLAR sign in row 10, the row number NOT varied. It remains 10.
  - There the formula produced in cell D2 is: A10





# Predicting The Number of Customers



# Predicting The Number of Customers At 10th Year



- **Given inputs**
  - Number of customers at the beginning of year 1 = 100
  - Number of customers acquired over the year = 20
  - Percentage of customers who leave the services = 15%
- **Problem is to compute the number of customers at the end of year 10.**
- **Solution approach is to compute the following four quantities**
  - Number of customers at the beginning of each year
  - New customers joined in that year
  - Estimated number of customers who quit the services
  - Customers at the end of that year

# Predicting The Number of Customers At End of 1st Year



|                                                            |                                         |
|------------------------------------------------------------|-----------------------------------------|
| <b>Number of customers at the beginning 1st year</b>       | <b>100</b>                              |
| <b>New customers joined in 1st year</b>                    | <b>20</b>                               |
| <b>Estimated number of customers who quit the services</b> | <b>15% of 100 = 15</b>                  |
| <b>Customers at the end of that year</b>                   | <b><math>100 + 20 - 15 = 105</math></b> |

# Predicting The Number of Customers At End of 2nd Year



|                                                            |                                               |
|------------------------------------------------------------|-----------------------------------------------|
| <b>Number of customers at the beginning 2nd year</b>       | <b>105</b>                                    |
| <b>New customers joined in 2nd year</b>                    | <b>20</b>                                     |
| <b>Estimated number of customers who quit the services</b> | <b>15% of 105 = 15.75</b>                     |
| <b>Customers at the end of that year</b>                   | <b><math>105 + 20 - 15.75 = 109.25</math></b> |

# Predicting The Number of Customers At End of 3rd Year



|                                                            |                                                     |
|------------------------------------------------------------|-----------------------------------------------------|
| <b>Number of customers at the beginning 3rd year</b>       | <b>109.25</b>                                       |
| <b>New customers joined in 3rd year</b>                    | <b>20</b>                                           |
| <b>Estimated number of customers who quit the services</b> | <b>15% of 109.25 =<br/>16.388</b>                   |
| <b>Customers at the end of that year</b>                   | <b><math>109.25 + 20 - 16.388 = 112.8625</math></b> |





# **Analysis - Demand Curve Estimation**





# Coffee Demand Curve Estimation

- **Given inputs**
  - **Coffee Demand (in number of cups) =  $(100 - 15 * \text{price})$**
  - **Profit =  $(\text{Price of cup} - \text{Original unit cost}) * \text{Coffee Demand}$**
- **Estimate the profit when**
- **The original unit cost varies from \$0.5 to \$2.0 and**
  - **Price of one coffee cup is sold is varied from \$2.0 to \$5.0.**
- **Solution approach is to compute the following four quantities**
  - **Number of customers at the beginning of each year**
  - **New customers joined in that year**
  - **Estimated number of customers who quit the services**
  - **Customers at the end of that year**

# Coffee Demand Curve Estimation - Example



| Unit Cost | Price | Demand              | Profit                 |
|-----------|-------|---------------------|------------------------|
| 0.5       | 2.0   | $100 - 15 * 2 = 70$ | $(2 - 0.5) * 70 = 105$ |
| 1.0       | 2.0   | $100 - 15 * 2 = 70$ | $(2 - 1.0) * 70 = 70$  |
| 1.5       | 2.0   | $100 - 15 * 2 = 70$ | $(2 - 1.5) * 70 = 35$  |
| 2.0       | 2.0   | $100 - 15 * 2 = 70$ | $(2 - 2) * 70 = 0$     |

# Coffee Demand Curve Estimation - Example



| Unit Cost | Price | Demand                       | Profit                            |
|-----------|-------|------------------------------|-----------------------------------|
| 0.5       | 2.5   | $100 - 15 \times 2.5 = 62.5$ | $(2.5 - 0.5) \times 62.5 = 125$   |
| 1.0       | 2.5   | $100 - 15 \times 2.5 = 62.5$ | $(2.5 - 1.0) \times 62.5 = 93.75$ |
| 1.5       | 2.5   | $100 - 15 \times 2.5 = 62.5$ | $(2.5 - 1.5) \times 62.5 = 62.5$  |
| 2.0       | 2.5   | $100 - 15 \times 2.5 = 62.5$ | $(2.5 - 2) \times 62.5 = 31.25$   |





## **Naming Group of Cells – Range Name**





# Range Names

- Cells are referred using (Column Name Row Number) notation
- A1, B7, D12 etc
- Group of cells are referred using colon notation
  - Rows 1 to 100: A1:A100
  - Columns B to G: B7:G7
  - Rows & Columns: A1:C3 (A1, A2, A3, B1, B2, B3, C1, C2 and C3)
- It is easier if cells are given name instead of A1, B7, D12 or name to group of cells such as marks instead of A1:A100
- Name to cells are given in the interface Name box



**Name individual cells**





## **Example – 01 – State-wise Sales**



## **Example – 02 – Stocks, TBills and Bonds**





## **Example – 03 – Revenue Growth- Referring Previous Cell**





## Example – 04 – Wages







**Thank You!**

